PROTOTYPE · THE CHANGE SENTINEL

Find new stone quarries from orbit. Hand officers case files, not curiosities.

A laptop pipeline that watches Nuh district, Haryana, in free Sentinel-2 imagery, filters out clouds and seasons, and ranks every new quarry it finds with the evidence attached.

Sentinel-2 L2A, 10–20 m

Python, one laptop

No accounts or API keys



NUH-2609-0dc7e1, Ghata Shamasabad(106), Ferozepur Jhirka. Sentinel-2 true colour, 30 Sep 2026; outline is the detected change.

THE PROBLEM

Illegal quarrying is visible from orbit long before anyone complains, but nobody is watching continuously

2009

Supreme Court suspends all mining in the Aravalli hills of Faridabad, Gurugram and Nuh. Illegal pits keep appearing.

14%

of national Mining Surveillance System alerts were confirmed as illegal mining by the states (RTI response).

Manual comparison doesn't scale

Flicking between images works for one village. No district office does it every month for 1,800 km².

Naive change detection floods

Clouds, shadows, monsoon green-up and harvests produce thousands of false alarms.

Commercial monitoring is out of reach

Priced for corporate customers, not a district office or a citizen group.

Alerts aren't actionable

No jurisdiction, no area, no before-and-after view. An officer can't act on a dot on a map.

OUR CHOICE

One change class: new open-cast stone quarries in Nuh's Aravalli hills

1

"Unauthorised" can be decided

A quarry is inside a lease polygon or it isn't. In Nuh's hills all mining has been suspended since 2009, so a new pit there is presumptively illegal.

2

The signal survives the seasons

A field is bare for a few months, then green again. A working quarry is bare rock or a flooded pit in nearly every clear view, all year.

3

It fixes the existing failure

Alerts without evidence and ranking get ignored. Every case here ships with images, area, jurisdiction and a written reason for its rank.

A PIXEL IS FLAGGED WHEN

≥ 75%
of this year's clear views show bare ground or an open pit

+30 points
more often bare than in the same 12 months a year earlier

−0.10
drop in peak greenness (90th-percentile NDVI)

≥ 0.5 ha
after speckle removal, as one connected site

ARCHITECTURE

From scene search to case file in eight steps

1 District and grid

geoBoundaries ADM1–5 → district mask plus 1 km buffer on a 20 m UTM grid

`aoi.py`

→

2 Scene search

Earth Search STAC, Sentinel-2 L2A Collection 1. The 3 clearest dates per month

`imagery.py`

→

3 Yearly composites

One date at a time: cloud mask, bare-view share, peak and monthly NDVI

`composites.py`

→

4 Context layers

ESA WorldCover cropland share, Copernicus DEM slope

`ancillary.py`

5 Detection

Pixel rule → clean-up → sites ≥ 0.5 ha → polygons, "new" or "expansion"

`detect.py`

→

6 Jurisdiction

Village, block, tehsil, district, state. Lease check, border flag

`jurisdiction.py`

→

7 Ranking

Confidence → tier → priority → plain-language reasons

`prioritise.py`

→

8 Case files

Before/after, NDVI chart, onset month, hashes → HTML, GeoJSON, CSV

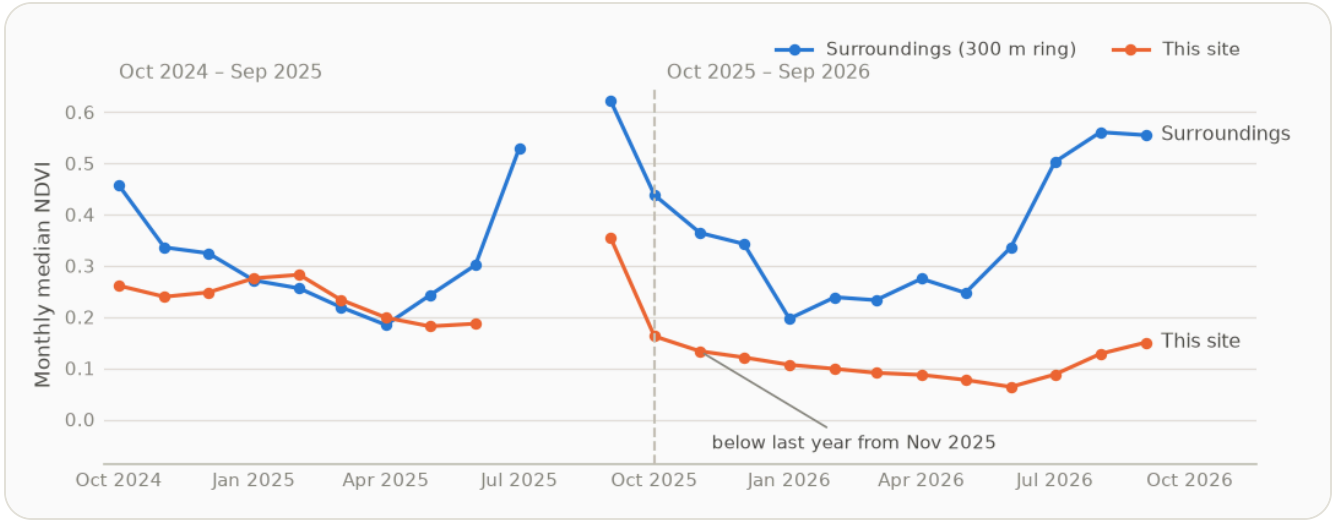
`evidence.py · report.py`

FREE DATA

- Sentinel-2 L2A C1
- Copernicus DEM GLO-30
- ESA WorldCover 2021
- geoBoundaries ADM1–5 (ODbL)

Clouds and seasons cancel out by design, not by tuning

- **Cloud and shadow mask** from the scene classification layer, grown by 40 m.
- **Peak greenness uses the 90th percentile.** A missed cloud can only lower NDVI, never raise it.
- **Whole 12-month windows**, so every season appears once on both sides of the comparison.
- **Onset dated against the same month last year**, relative to a 300 m ring around the site, so a wet or dry year cancels out too.

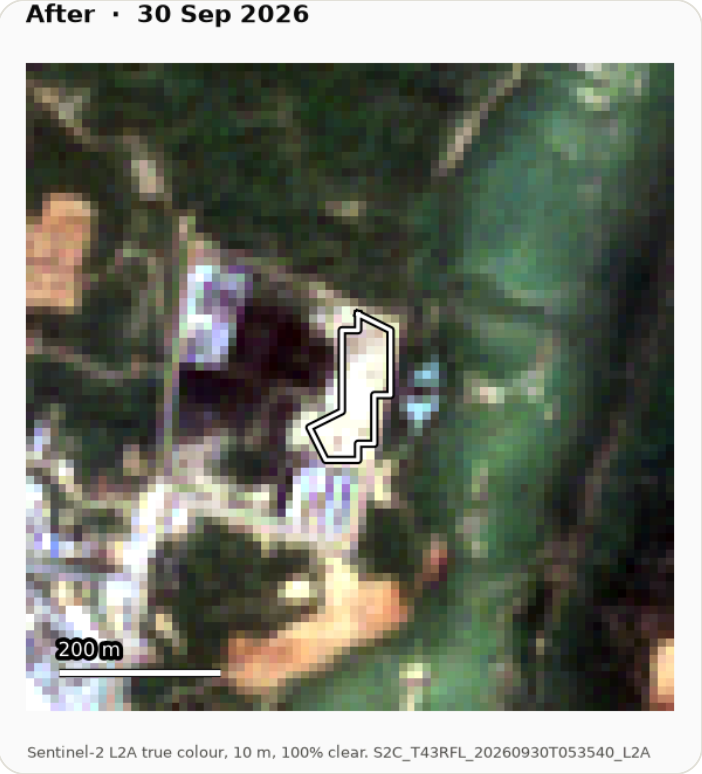


NUH-2609-0dc7e1: monthly NDVI inside the site against a 300 m ring. Seasons move both lines; the quarry pulls the site line down and keeps it there.

THE OUTPUT

Every detection becomes a case file an officer can act on





NUH-2609-0dc7e1: same-season pair, identical colour stretch, detected area outlined. Scene IDs and SHA-256 of each image are recorded in the case file.

VILLAGE
Ghata Shamasabad(106)

TEHSIL, DISTRICT
Ferozepur Jhirka, Nuh

AREA
1.0 ha (0.4–1.5)

CONFIDENCE
73% · Inspect

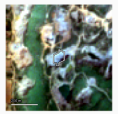
CHANGE WINDOW
2025-09-25 to 2026-09-30, onset Nov 2025

- Plain-language reasons for its rank
- Authorisation status and the legal note
- Monthly NDVI chart, site vs surroundings
- Field checklist and links to Google Maps / OSM
- evidence.json with geometry, STAC links, hashes

PRIORITISATION

The top of the list is worth an officer's Monday morning

Ranked cases

#	PRIORITY	CASE	LOCATION	AREA (HA)	CONFIDENCE	CHANGE HAPPENED BETWEEN	AUTHORISATION	AFTER
1	Inspect	NUH-2609-0dc7e1 expansion	**Ghata Shamasabad(106)** Ferozepur Jhirka, Ferozepur Jhirka	1.0 0.4–1.5	73%	2025-09-25 and 2026-09-30 onset Nov 2025	no authorised areas on file	
2	Inspect	NUH-2609-eb71c2 expansion	**Chinawara** Pahari, Pahari	1.0 0.6–1.5	93%	2024-10-05 and 2026-09-30	no authorised areas on file	
3	Inspect	NUH-2609-cb39ab expansion	**Chinawara** Pahari, Pahari	1.0 0.4–1.6	82%	2024-10-05 and 2026-09-30	no authorised areas on file	
4	Inspect	NUH-2609-7901cd expansion	**Chinawara** Pahari, Pahari	0.6 0.3–1.0	93%	2024-10-20 and 2026-09-30	no authorised areas on file	

The ranked index from the test run: tier, case, village, area, confidence, change window, thumbnail.

- **Inspect** confidence ≥ 70%: send someone
- **Review** 40–70%: check the images first
- **Low** below 40%: likely a fallow field

$$\text{priority} = \text{confidence} \times \log_2(1 + \text{hectares})$$

- ×0.1 inside an authorised lease, ×0.5 outside the district
- Confidence weighs bare share, its increase, NDVI drop, observation count, slope, and cropland share
- Stable case IDs: a re-run never renumbers an officer's list

RESULTS SO FAR

Test run: 9 sites in 211 km², 5 worth sending someone to

211

km² watched, Ferozepur Jhirka tehsil and the Rajasthan border

68

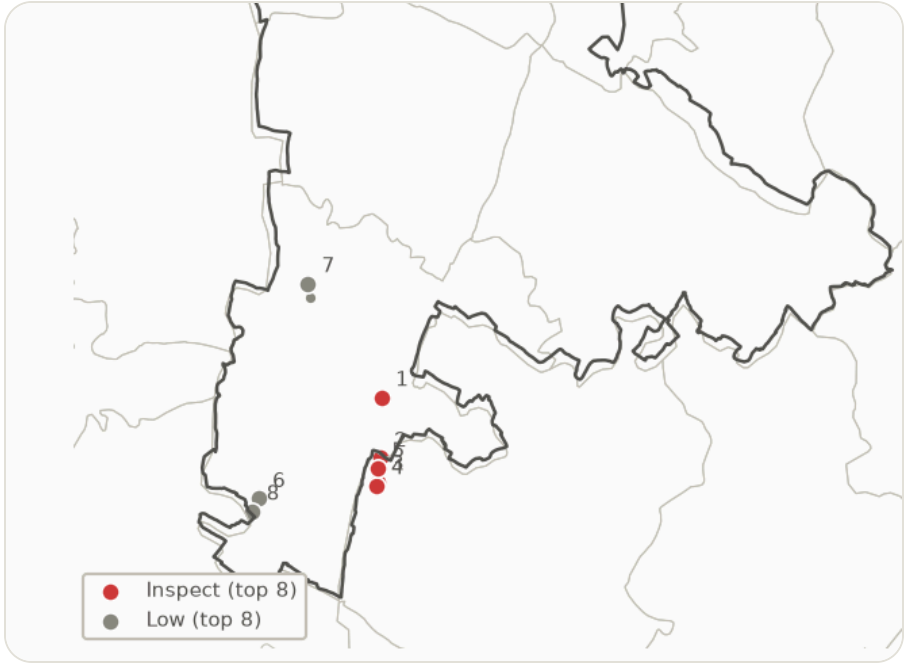
Sentinel-2 dates read across two 12-month windows

9

sites of 0.5 ha or more flagged by the rule and kept on the list

5 Inspect **4 Low**

- 4 sites just across the Rajasthan border, in an active quarry cluster, are kept and marked "refer to the neighbouring administration".
- 4 on mapped cropland, likely fallow fields, are kept but ranked last.
- Caught in testing: a network drop silently blanked six months of imagery. Downloads now retry, then stop the run rather than cache bad data.
- The whole district (~6 GB of imagery) has not been run yet: the link to the imagery servers is the limit, not the code.



Nuh district with the test area's cases, numbered by rank.

COMPUTE DISCIPLINE

Runs on one laptop. The bottleneck is bytes downloaded, not CPU

	TEST AREA	WHOLE DISTRICT
Area	211 km²	1,848 km²
Image dates	68	~68
Download	~0.5 GB	~6 GB (est.)
Peak memory	< 1 GB	~1.5 GB
CPU time	~1 min	a few minutes
First run, wall time	~20 min, slow link	1 h+ at 10 Mbps (est.)
Re-run, new rules	composites cached	composites cached

- **Each image block is downloaded once:** a whole date is read in one call. Tiling cost 4× the bytes in testing.
- **3 clearest dates per month.** Seasonal spread matters more than scene count.
- **10 m bands read from their 20 m overviews.** Only the evidence images use 10 m.
- **1-byte quantised NDVI stacks:** a year of observations fits in under 1 GB.
- **Cached composites:** change thresholds and re-run without downloading the year of imagery again.

HONEST LIMITS AND NEXT STEPS

A lead, not proof. Here is what it can't do yet

Confidence is not calibrated

Weights are hand-set priors. Next: label ~100 sites, fit the logistic model, report precision at "Inspect".

Monsoon gap

Optical imagery is thin July–September. Next: Sentinel-1 or NISAR radar, both free.

Fallow fields and brick kilns

Can stay bare all year. Cropland and slope push them down the list but don't remove them.

No lease layer for Nuh's neighbours

Next: load PARIVESH or state mining lease polygons; the lease check already supports them.

Whole district, every month

Run on a cloud notebook near the imagery, then schedule monthly so new cases appear automatically.

Other districts and change classes

A new YAML file per district. Lake infill is the inverted rule: water share falls, bare or built rises.

TRY IT

Open source, free data, one command

```
# install
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt

# test area (~20 min first run on a slow link)
.venv/bin/python -m sentinel config/nuh.yaml \
    --subset 76.86,27.70,77.00,27.86

# whole district
.venv/bin/python -m sentinel config/nuh.yaml
open output/nuh/index.html
```

LANGUAGE	Python 3.12
IMAGERY	pystac-client, odc-stac, odc-geo, rasterio / GDAL
PROCESSING	numpy, scipy.ndimage, xarray, pandas
MAPS	geopandas, shapely, pyproj
CASE FILES	matplotlib, Jinja2 → HTML, GeoJSON, CSV, JSON
INFRASTRUCTURE	A laptop. No server, database, account or key
TESTS	pytest: detection, dating and download rules